
Week 4 – Progress Report

SERGE: AI Agent for Supply Chain Operations

Use Case Definition, Specifications & Evaluation Design

Louis Grignola Gauthier Schmitt
Maël Tellus Matteo Schwarz

Course: SE618-1 Logistics

Institution: Sirindhorn International Institute of Technology (SIIT)

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Advisor Feedback & Project Recentering

Following our Week 3 advisor meeting, several action items were identified to sharpen the project scope before moving to implementation. The feedback highlighted the need to:

1. **Define a clear field:** choose between B2B and B2C, and narrow the operational context.
2. **Identify a real customer pain point:** ground the project in a concrete, observable problem.
3. **Clarify the agent’s role:** is it predictive problem resolution, data retrieval, or diagnostic reasoning?
4. **Design evaluation with ground truth:** use an LLM-as-Judge approach with well-defined expected answers.
5. **Add proper references:** strengthen the methodology report with citations from our literature review.

This week’s deliverable addresses all five points by defining the project specifications: who the user is, what problem we solve, how we evaluate it, and what success looks like.

Project Positioning

Field: B2B Distribution

We position SERGE in a **B2B mid-size distribution company** context. This type of company purchases goods from suppliers, stores them in warehouses, and ships them to business customers (retailers, manufacturers, or other distributors).

Key Insight

B2B distribution is the ideal setting because it involves all four supply chain systems (SRM, WMS, OMS, TMS) with high interdependency. A single delayed supplier shipment can cascade into stockouts, delayed customer orders, and carrier rebooking — exactly the kind of cross-system reasoning our agent targets.

Why not B2C? In B2C (e-commerce, retail), the end customer typically interacts with a single storefront. The operational complexity we target — cross-system investigation across procurement, warehousing, order management, and transport — is a B2B operations team problem, not a consumer-facing one.

Agent Role: Data Retrieval & Diagnostic Reasoning

We explicitly scope SERGE as a **retrieval and diagnostic** agent, not a predictive or decision-making one:

Capability	In Scope?	Description
Data retrieval	Yes	Query any of the 4 systems via natural language
Cross-system diagnostic	Yes	Connect data across SRM, WMS, OMS, TMS to find root causes
Report generation	Yes	Produce structured summaries and comparisons
Predictive analytics	No	Forecasting demand, predicting delays (future work)
Autonomous decisions	No	Reordering stock, rerouting shipments (out of scope)

This scoping is deliberate: in practice, companies deploy read-only AI assistants *before* trusting AI with write access [1, 2]. Our agent reflects this real-world adoption pattern.

Customer Pain Point

Customer Pain Point

Problem: In a B2B distribution company, operations managers spend **15–30 minutes per investigation** when diagnosing supply chain issues (delayed orders, stockouts, supplier problems). They must manually log into 3–4 separate systems, extract data from each, and mentally piece together the causal chain.

Consequences:

- Slow response to customer inquiries about order status
- Incomplete investigations (human misses a data point in one system)
- Knowledge concentrated in senior staff who know “where to look”
- No structured audit trail of investigations

User Persona

Attribute	Description
Role	Operations Manager / Supply Chain Coordinator
Company	Mid-size B2B distributor (50–500 employees)
Daily tasks	Monitor order fulfillment, investigate delays, coordinate with suppliers and carriers, report on KPIs
Systems used	SRM, WMS, OMS, TMS (often separate vendors, separate logins)
Pain	Spends 2–3 hours/day on cross-system lookups; relies on experience to know which system to check first
Goal	Get answers in seconds, not minutes; never miss a data point

Value Proposition

Dimension	Without SERGE	With SERGE
Investigation time	15–30 min (manual, 3–4 systems)	<1 min (single natural language query)
Completeness	Depends on operator experience	Systematic: always checks all relevant systems
Knowledge dependency	Senior staff bottleneck	Any team member can investigate
Audit trail	None (mental process)	Structured report with sources cited

Use Case Specifications

We define **three use cases**, each representing a common operations task. They are deliberately simple and concrete to allow rigorous evaluation.

UC-1: Order Status Inquiry

Title	Order Status Inquiry
Actor	Operations Manager
Trigger	A customer calls asking about their order
Question	<i>“What is the status of order #ORD-2024-0456?”</i>
Systems	OMS (order details) → TMS (shipment tracking)
Expected flow	1. Agent queries OMS for order details (status, items, dates) 2. Agent queries TMS for associated shipment(s) 3. Agent returns a unified status: order info + shipping status + ETA
Success	Correct order data, correct shipment status, correct ETA
Complexity	Low — 2 systems, direct lookup

Why this use case?

This is the most frequent query in any distribution company. It validates the agent’s basic ability to retrieve and combine data from two systems. If the agent cannot do this reliably, nothing else matters.

UC-2: Delayed Order Root Cause Analysis

Title	Delayed Order Root Cause Analysis
Actor	Operations Manager
Trigger	An order is overdue and the manager needs to explain why
Question	<i>“Why is order #ORD-2024-0456 delayed?”</i>
Systems	OMS → TMS → WMS → SRM (full chain)
Expected flow	1. Agent queries OMS: order is overdue, status = processing 2. Agent queries TMS: no shipment created yet 3. Agent queries WMS: insufficient stock for one item 4. Agent queries SRM: purchase order exists but is delayed from supplier 5. Agent synthesizes: “Order delayed because SKU-X is out of stock. A purchase order is pending from Supplier Y, expected in 3 days.”
Success	Correctly identifies the root cause (inventory shortage → supplier delay)
Complexity	Medium — 4 systems, causal chain reasoning

Why this use case?

This is the core value proposition of SERGE. A human doing this manually would need to check 4 systems sequentially, each time using the output of the previous query as input for the next. This is exactly the kind of cross-system reasoning that justifies an agentic approach over a simple chatbot.

UC-3: Supplier Performance Comparison

Title	Supplier Performance Comparison
Actor	Operations Manager / Procurement
Trigger	Need to decide which supplier to use for a new purchase order
Question	<i>“Compare the performance of our suppliers for electronic components”</i>
Systems	SRM (supplier data, PO history, performance metrics)
Expected flow	1. Agent queries SRM for suppliers in the relevant product category 2. Agent queries SRM for performance metrics (on-time rate, lead time, defect rate) 3. Agent queries SRM for recent PO history 4. Agent generates a comparison table with metrics
Success	Correct metrics for each supplier, accurate comparison, reasonable ranking
Complexity	Low–Medium — 1 system, multiple queries, structured output

Why this use case?

This demonstrates the agent’s ability to perform structured data extraction and present it in a useful format (comparison table). Unlike UC-1 and UC-2 which are investigative, this is an *analytical* query that tests report generation quality.

Use Case Summary

UC	Name	Systems	Complexity	Tests
1	Order Status Inquiry	OMS, TMS	Low	Retrieval accuracy, data merging
2	Delayed Order Root Cause	OMS, TMS, WMS, SRM	Medium	Cross-system reasoning, causal chain
3	Supplier Performance	SRM	Low–Med	Structured output, data analysis

Evaluation Design: LLM-as-Judge

Approach

Following our advisor’s recommendation, we adopt an **LLM-as-Judge** evaluation methodology. This requires:

1. **Ground truth:** For each test scenario, we predefine the correct answer based on the seed data. We know exactly what the agent *should* find.
2. **Agent execution:** The agent processes the query and produces a response.
3. **Judge evaluation:** A separate LLM (the “judge”) compares the agent’s response against the ground truth and scores it on predefined criteria.

Key Insight

The LLM-as-Judge approach is well-suited for our project because our agent’s outputs are *natural language reports*, not numerical predictions. Traditional metrics (accuracy, F1) don’t capture whether the agent correctly explained a root cause. A judge LLM can assess both factual correctness and reasoning quality.

Ground Truth Design

For each use case, we define **test scenarios** with known answers embedded in the seed data.

Scenario 1: Order Status (UC-1)

Input	“What is the status of order #ORD-2024-0456?”
Ground truth	• Order #ORD-2024-0456: placed 2024-01-10, customer = MegaCorp, status = shipped • Contains 2 line items: 50x SKU-WIDGET-A, 20x SKU-GADGET-B • Shipment SHP-2024-1892: carrier = DHL, status = in transit, ETA = 2024-01-17
Must contain	Order ID, customer name, order status, item list, shipment ID, carrier, ETA
Must NOT contain	Fabricated data, incorrect quantities, wrong carrier

Scenario 2: Root Cause Analysis (UC-2)

Input	“Why is order #ORD-2024-0789 delayed?”
Ground truth	- Order #ORD-2024-0789: required by 2024-01-15, status = processing (not shipped) - No shipment created yet - WMS: SKU-MOTOR-X has 12 units available, order requires 50 - SRM: PO-2024-102 for 150 units of SKU-MOTOR-X from TechParts Inc, status = in transit, ETA = 2024-01-18 - Root cause: inventory shortage due to pending supplier delivery
Must contain	Inventory shortage identified, supplier PO referenced, expected resolution date
Must NOT contain	Incorrect root cause, missing causal link

Scenario 3: Supplier Comparison (UC-3)

Input	“Compare our suppliers for electronic components”
Ground truth	3 suppliers in category: Acme Electronics, TechParts Inc, GlobalSupply Co - On-time rates: Acme 91%, TechParts 96%, GlobalSupply 78% - Avg lead times: Acme 12 days, TechParts 8 days, GlobalSupply 15 days - TechParts has best reliability; GlobalSupply has lowest cost but worst reliability
Must contain	All 3 suppliers listed, on-time rates, lead times, comparative analysis
Must NOT contain	Missing suppliers, fabricated metrics

Judge Evaluation Criteria

The judge LLM scores each agent response on 5 dimensions (1–5 scale):

Criterion	Weight	What the Judge Assesses
Factual correctness	30%	Are all stated facts accurate compared to ground truth?
Completeness	25%	Does the response cover all required data points?
Reasoning quality	20%	Is the causal chain logical and well-explained? (UC-2 mainly)
Source attribution	15%	Does the agent cite which system each piece of data comes from?
Response structure	10%	Is the output well-organized and easy to read?

Evaluation Protocol

1. Prepare **5 scenarios per use case** (15 total) with ground truth
2. Run each scenario **3 times** to measure consistency (45 runs total)
3. Judge LLM scores each response independently
4. Compute per-use-case and aggregate scores
5. Also measure: number of MCP tool calls, total response time

Metric	Type	Target	How Measured
Judge score	Quality	$\geq 4.0/5.0$	LLM-as-Judge weighted average
Factual accuracy	Correctness	$\geq 90\%$	Judge criterion #1
Scenario pass rate	Reliability	$\geq 80\%$	% of runs scoring $\geq 3.5/5.0$
Avg tool calls	Efficiency	≤ 6 per query	Count of MCP calls per scenario
Response time	Performance	< 30 seconds	End-to-end wall clock time
Consistency	Stability	$\sigma \leq 0.5$	Std deviation across 3 runs

Table 1: Quantitative evaluation targets.

Objectives Summary

Question	Answer
Who is the user?	Operations manager at a mid-size B2B distributor
What is the pain?	Cross-system investigations take 15–30 min because data is siloed across SRM, WMS, OMS, TMS
What does SERGE do?	Retrieves data from all 4 systems via natural language and produces diagnostic reports in <1 min
What is the scope?	Read-only retrieval + diagnostic reasoning. No predictions, no write operations
Why agentic AI?	A simple chatbot cannot chain queries across systems adaptively. An agent reasons about <i>what to query next</i> based on intermediate results (e.g., “stock is low → check supplier POs”)
How do we evaluate?	LLM-as-Judge with predefined ground truth; 15 scenarios, 45 runs
What does success look like?	Judge score $\geq 4.0/5.0$, factual accuracy $\geq 90\%$, response time <30s

Updated References

The following references support the design decisions made in this report. Numbers [1]–[14] refer to our Week 3 literature catalogue; new references are marked with *.

- [1] Jannelli, V. et al. (2024). Agentic LLMs in the Supply Chain: Towards Autonomous Multi-Agent Consensus-Seeking. *arXiv:2411.10184*. *Int. J. Prod. Res.*, 2025.
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- [3] AlMahri, S. et al. (2026). Automating Supply Chain Disruption Monitoring via an Agentic AI Approach. *arXiv:2601.09680*. — *Supports our synthetic scenario evaluation design (Section 5)*.
- [4] Yoshizato, K. et al. (2026). AI Agent Systems for Supply Chains: Structured Decision Prompts & Memory Retrieval. *arXiv:2602.05524*. AAMAS 2026. — *Informs our multi-step reasoning approach in UC-2*.
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- [7] Zhang, Y. et al. (2025). Multi-Agent RAG for Supply Chain Knowledge Bases. *arXiv:2506.17484*. KDD 2025 Workshop. — *Validates our approach of querying structured SC data via an agent*.
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- [11] Li, B. et al. (2023). OptiGuide: LLMs for Supply Chain Optimization. *arXiv:2307.03875*.
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- [15]* Zheng, L. et al. (2023). Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena. *NeurIPS 2023*. *arXiv:2306.05685*. — *Foundational paper for our LLM-as-Judge evaluation methodology*.
- [16]* Shankar, S. et al. (2024). Who Validates the Validators? Aligning LLM-Assisted Evaluation with Human Preferences. *arXiv:2404.12272*. — *Guidelines for designing reliable judge prompts and scoring rubrics*.

Next Steps

- **Week 5:** Implement PostgreSQL schema for all 4 systems; write seed data scripts with the 15 evaluation scenarios embedded as ground truth.
- **Week 5–6:** Build MCP servers (SRM, WMS first, then OMS, TMS) with read-only tools.
- **Week 6:** Implement the LLM-as-Judge evaluation pipeline; design judge prompt and scoring rubric.
- **Week 7:** Agent integration, run evaluation scenarios, iterate on agent prompts.